AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2019

Chemistry Paper II

Time: 2 hours 15 minutes Marks: 55

April/ May 2019
for
Teaching & Learning only

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are ELEVEN questions. Answer ALL questions. Questions 10 & 11 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 5 Marks)

a.

i. Which of the following metals has higher ionisation energy? Give reason to support your answer. (2 Marks)

I. Caesium

II. Lithium

ii. Why is the radius of fluorine atom smaller than that of the fluoride ion? (1 Mark)

b. Write the products formed when the given nitrate of lithium (Li) is decomposed. (2 Marks)

$4LiNO_3 \longrightarrow$ _____

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Q.2. (Total 5 Marks)

Write the systematic names of the given transition metal complexes.

Transition Metal Complex	Systematic Name
$Na_3[FeCl(CN)_5]$	
$[Pt(NH_3)_2Cl_2]Cl_2$	
$Na_2[NiCl_4]$	
$K_3[CoF_6]$	
$[Co(NH_3)_4Cl_2]Cl$	

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Q.3. (Total 4 Marks)

a. Draw the following structures with the molecular formula $C_7H_{15}Br$.

i. A secondary alkyl halide (1 Mark)

Space for drawing

ii. A tertiary alkyl halide (1 Mark)

Space for drawing

b. 2-bromo-2-methylpropane in the presence of alcoholic potassium hydroxide forms 2-methylpropene.

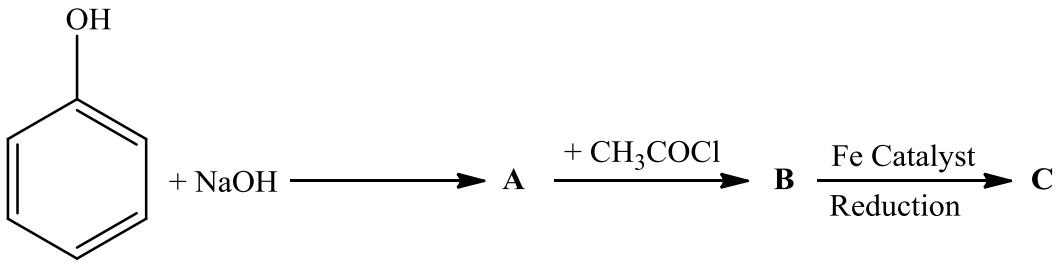
Deduce the mechanism of the given chemical reaction. (2 Marks)

Space for deduction of mechanism

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Q.4. (Total 5 Marks)

a. Draw the structures of **A**, **B** and **C** to complete the given reaction of phenol. (3 Marks)



Space for drawing

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b. Why is iodoform test used to differentiate between methanol and ethanol? Give reason with reference to the structure of methanol and ethanol. (2 Marks)

Q.5. (Total 4 Marks)

Show the preparation of acetic acid by reacting methylmagnesium bromide (Grignard's reagent) with carbon dioxide. Mention the conditions necessary for this reaction.

Space to show reaction and conditions

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Q.6. (Total 4 Marks)

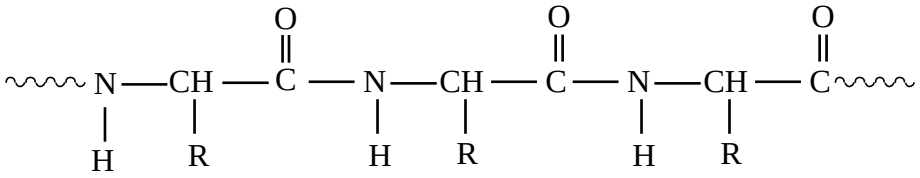
Draw the structural formulae of the following compounds.

Compound	Structural Formula
Propanone phenylhydrazone	
4-hydroxybenzaldehyde	
Ethanaloxime	
3-Methyl-2-butanone	

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Q.7. (Total 4 Marks)

A section of protein is shown below.



i. Circle the peptide linkage in the given section of protein. (1 Mark)

ii. Draw the structural formula for the monomer used to make the given protein. (1 Mark)

Space for drawing

iii. Describe the type of polymerisation process involved in the formation of the given protein. (2 Marks)

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Q.8. (Total 5 Marks)

a. Describe the process of destructive distillation of coal and name any ONE product obtained through this process. (2 Marks)

b.

i. What causes industrial and photochemical smog? (1 Mark)

ii. Mention any TWO harmful effects of smog on living organisms. (2 Marks)

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Q.9. (Total 4 Marks)

a. Infrared (IR) and proton nuclear magnetic radiation (^{1}H-NMR) spectroscopy are both used to identify structures of molecules.

Which specific type of information does each provide? (2 Marks)

i. IR spectroscopy:

_____

_____

ii. ^{1}H-NMR spectroscopy:

_____

_____

b. State ONE use and ONE drawback of combustion analysis. (2 Marks)

_____

_____

_____

_____

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Q.10. (Total 8 Marks)

EITHER

a.

i. Explain why alkenes are more reactive towards electrophilic reagents than alkynes.(4 Marks)

ii. Write a chemical equation showing preparation of alkenes from (4 Marks)

I. dehydrohalogenation of alkyl halide.

II. dehydration of alcohol.

(**Note**: In the equations, mention all the necessary conditions for reactions.)

OR

b. Using balanced chemical equations,

i. show the preparation of propanoyl chloride by reacting propanoic acid with thionyl chloride in the presence of pyridine. (2 Marks)

ii. show the reaction of propanoyl chloride with (6 Marks)

I. sodium propanoate to distill over propanoic anhydride.

II. propanol in the presence of pyridine to prepare propyl propionate.

III. ammonia to form propanamide.

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Q.11. (Total 7 Marks)

EITHER

a.

i. Why are enzymes called biological catalysts? (1 Mark)

ii. Explain the effects of the following factors on enzyme activity: (6 Marks)

I. Enzyme concentration

II. Temperature

III. pH

OR

b.

i. Write the importance of each of the given pesticides: (4 Marks)

I. Molluscicides

II. Rodenticides

III. Herbicides

IV. Fungicides

ii. What are dyes? (1 Mark)

iii. What is the colour of methyl orange in acidic medium and basic medium? (2 Marks)

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END OF PAPER

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